MIL-T-47066(MI)
10 May 1974
SUPERSEDING
MIS-14331A
12 June 1967

MILITARY SPECIFICATION

TITANIUM DIOXIDE, REAGENT GRADE

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one grade of titanium dioxide.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard
PPP-C-850	Cushioning Material Polystyrene, Expanded, Resilient (For Packaging Uses)
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (for carton sealing)

FSC 6810

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM E-2 SM 11-2	Suggested Method for Spectrochemical Analysis of Unknown Materials by Powder-D-C Arc Semiquentitative Technique

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample of titanium dioxide shall meet the requirements of this specification.

3.2 Material. The material shall be a high purity grade of titanium dioxide.

3.3 Impurities. The impurities of the titanium dioxide shall be limited to, and shall not be in excess of the maximum of Table I.

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Table I

Impurities	
Impurity	Percentage (maximum)
Water Soluble Salts	0.05
Arsenic (As)	0.0001
Iron (Fe)	0.010
Lead (Pb)	0.010
Zinc (Zn)	0.010

3.4 Workmanship. The workmanship shall be such as to insure a product which is uniform and in conformance with this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent lots of material. Preproduction samples which do not meet the requirements of this specification shall be subject to rejection and return to the supplier.

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4.3 Classification of tests. The inspection and testing of the titanium dioxide shall be classified as follows:

a. Preproduction tests.

b. Quality conformance tests.

4.3.1 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all the examinations and tests specified herein.

4.3.2 Quality conformance tests. Quality conformance tests for acceptance of the material shall consist of all of the examinations and tests of this specification and shall be performed by the supplier on a sample of the titanium dioxide representative of the shipment. A certified analysis shall be furnished with each lot.

4.3.2.1 Lot size. Lot size shall consist of all the titanium dioxide manufactured at one time from one batch, forming part of one contract or order, and submitted for acceptance at the same time and place.

4.3.2.2 Sampling. Unless otherwise specified (see 6.2), 100 grams from one unit, representative of each lot shall be selected at random for quality conformance testing. Each container in lot shall be considered as a unit of product. Failure of sample to meet the quality conformance test requirements shall be cause for lot rejection (see 4.5).

4.4 Test methods.

4.4.1 Chemical composition. The chemical composition shall be determined by spectrographic analysis in accordance with ASTM E-2 SM 11-2 or an approved substitute method.

4.5 Preservation, packaging, packing, and marking. The preservation, packaging, packing, and marking shall be examined for compliance with Section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified (see 6.2), preservation, packaging, packing, and marking shall be as specified herein.

5.1.1 Preservation and packaging.

5.1.1.1 Level A. One pound of titantium dioxide shall be packaged in screw cap bottles. The bottle shall be sealed with a secondary closure of self shrinking plastic film or conformable plastic tape. The containers shall be nonreactive to the contents. Units shall be cushioned with material conforming to PPP-C-850 and then inserted in containers conforming to PPP-B-636, Class 2. The material shall be packaged in such a manner that deterioration or physical damage during shipment and storage is prevented.

5.1.1.2 Level B. Same as Level A.

5.1.1.3 Level C. Unit package shall be in accordance with 5.1.1.1. Unit containers shall be in accordance with the suppliers commercial practice. Packages shall be protected against deterioration or physical damage during shipment from the supply source to the first receiving activity for immediate use.

5.1.2 Packing.

5.1.2.1 Level A. Unit packages or intermediate containers, packaged as prescribed in 5.1.1.1, shall be placed in a snug container conforming to PPP-B-636, Class 2. Closure shall be in accordance with the appendix of PPP-B-636 except that tape conforming to PPP-T-76 shall be used.

5.1.2.2 Level B. Unit packages or intermediate containers, packaged as prescribed in 5.1.1.1, shall be packed in containers conforming to PPP-B-636, Class 1. Closure shall be in accordance with the appendix of the applicable container specification.

5.1.2.3 Level C. Unit packages or intermediate containers, packaged as prescribed herein, shall be packed in

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a container that will ensure safe delivery at destination and in compliance with common carrier rules and regulations applicable to the mode of transportation.

5.1.3 Marking.

5.1.3.1 Bottles. Each bottle shall be labeled listing the following information:

a. Nomenclature.

b. Analysis or maximum impurities of contents.

c. Date of manufacture and lot number.

d. Stock number.

e. Quantity.

5.1.3.2 Packages. Each package shall have a permanent water-proofed label or tag firmly affixed thereto and listing the following information:

a. Nomenclature.

b. Stock number.

c. Quantity.

5.1.3.3 Shipping containers. In addition to any special marking required by the contract or purchase order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129 and shall include the following:

a. Title and number of this specification.

b. Manufacturer's name and address.

c. Manufacturer's product designation.

d. Date of manufacture.

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e. Contract number.

f. Quantity.

6. NOTES

6.1 Intended use. The titanium dioxide is intended for use as a coating for precision optical elements.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required and, if so, pertinent details (see 3.1).

c. Sampling plan if other than specified (see 4.3.2.2).

d. Applicable level of preservation, packaging, and packing (see 5.1.1 and 5.1.2).

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-14331A, dated 12 June 1967.

Custodian:
Army-MI

Preparing Activity:
Army-MI
Project No. 6810-AA13

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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